Software registration: SomnGuard / Registro de software: SomnGuard

SomnGuard | ADSO 3145555 | Revisión 2026-11-07

Fill in every empty field. It follows Decreto 1360 de 1989.

Llene todos los campos vacíos. Sigue el Decreto 1360 de 1989.

Institutional context / Marco institucional

Field / Campo	Value / Valor
Purpose / Propósito	Request registration before the Dirección Nacional de Derecho de Autor, for the apprentices team and the instructors team that supported the project or are cross-cutting to it / Solicitar el registro ante la Dirección Nacional de Derecho de Autor, a nombre del equipo de aprendices y del equipo de instructores que apoyaron el proyecto o son transversales a su consecución
Process / Proceso	Completion of the taught stage of the technologist program / Finalización de la etapa lectiva del tecnólogo
Training project / Proyecto formativo	Desarrollo de software orientado a servicios
Research seedbed / Semillero de investigación	Developers & Geeks

1. Applicant / Solicitante (Decreto 1360, art. 5, num. 1)

Field / Campo	Value / Valor
Full name / Nombre	Juan Carlos Jurado Castañeda
ID document / Identificación	CC 1077224670
Acts on own behalf or as representative / Actúa en nombre propio o como representante	[ ] Own behalf / Nombre propio [x] Representative / Representante de los demás cotitulares
Proof of representation / Prueba de representación, si aplica	According to the applicant, the co-holders gave verbal authorization for Juan Carlos Jurado Castañeda (CC 1077224670) to act as their representative. No written authorization is attached. / Según informa el solicitante, los cotitulares dieron autorización verbal a Juan Carlos Jurado Castañeda (CC 1077224670) para representarlos. No se adjunta autorización escrita.

2. Authors / Autores (art. 5, num. 2)

Authors are the instructors and the apprentices listed in `team-roster.md`. Each apprentice checks their data and writes their contribution. / Los autores son los instructores y aprendices de `team-roster.md`. Cada aprendiz revisa sus datos y escribe su aporte.

Full name / Nombre completo	Doc. type / Tipo	Doc. number / Número	Ficha	Contribution / Aporte
JESUS ARIEL GONZALEZ BONILLA		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
CARLOS FABIAN MARTINEZ MORA		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
CARLOS JULIO CADENA SARASTY		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
JAVIER HUMBERTO PINTO DIAZ		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
KAROL DANIELA CORREA TRUJILLO		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
JOSE DE JESUS MOTTA VARGAS		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
FERNELLY CARVAJAL SILVA		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
EDDIER YAIR LIBERATO CUENCA		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
JHON WILLIAN CORREDOR ARAUJO		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
ANDRES MORENO COLLAZOS		Kept privately / En custodia		Instructor: see `team-roster.md` / ver `team-roster.md`
JUAN CARLOS JURADO CASTAÑEDA	CC	1077224670	3145555	Programó y desarrolló el backend, construyó interfaces y funcionalidades del frontend, diseñó, creó e implementó la base de datos y participó en la integración de estos componentes para garantizar el funcionamiento del sistema.
JOHAN STEVEN RODRIGUEZ CHARRY	CC	1075794019	3145555	Programó y desarrolló el backend, construyó interfaces y funcionalidades del frontend, y diseñó, creó e implementó la base de datos del sistema.
CRISTIAN JAVIER PALMA SOTTO	CC	1141115893	3145555	Programó y desarrolló el backend, construyó interfaces y funcionalidades del frontend, diseñó y gestionó la base de datos, y elaboró y organizó la documentación técnica del proyecto.
BRAYAN ALBERTO PERDOMO PERDOMO	CC	1075235870	3145555	Desarrolló el frontend y realizó el diseño UX/UI.

3. Producer / Productor (art. 5, num. 3)

Field / Campo	Value / Valor
Producer name / Nombre del productor	Supporting and cross-cutting instructors / Instructores de apoyo y transversales: JESUS ARIEL GONZALEZ BONILLA, CARLOS FABIAN MARTINEZ MORA, CARLOS JULIO CADENA SARASTY, JAVIER HUMBERTO PINTO DIAZ, KAROL DANIELA CORREA TRUJILLO, JOSE DE JESUS MOTTA VARGAS, FERNELLY CARVAJAL SILVA, EDDIER YAIR LIBERATO CUENCA, JHON WILLIAN CORREDOR ARAUJO, ANDRES MORENO COLLAZOS. Project apprentices / Aprendices del equipo: JUAN CARLOS JURADO CASTAÑEDA, JOHAN STEVEN RODRIGUEZ CHARRY, CRISTIAN JAVIER PALMA SOTTO, BRAYAN ALBERTO PERDOMO PERDOMO.
ID document or NIT / Identificación o NIT	Apprentices: section 2. Instructors: kept privately by the lead instructor. / Aprendices: sección 2. Instructores: en custodia del instructor líder.

4. Rights holder / Titular de los derechos patrimoniales

Field / Campo	Value / Valor
Rights holder / Titular	Supporting and cross-cutting instructors / Instructores de apoyo y transversales: JESUS ARIEL GONZALEZ BONILLA, CARLOS FABIAN MARTINEZ MORA, CARLOS JULIO CADENA SARASTY, JAVIER HUMBERTO PINTO DIAZ, KAROL DANIELA CORREA TRUJILLO, JOSE DE JESUS MOTTA VARGAS, FERNELLY CARVAJAL SILVA, EDDIER YAIR LIBERATO CUENCA, JHON WILLIAN CORREDOR ARAUJO, ANDRES MORENO COLLAZOS. Project apprentices / Aprendices del equipo: JUAN CARLOS JURADO CASTAÑEDA, JOHAN STEVEN RODRIGUEZ CHARRY, CRISTIAN JAVIER PALMA SOTTO, BRAYAN ALBERTO PERDOMO PERDOMO.
ID document or NIT / Identificación o NIT	Apprentices: section 2. Instructors: kept privately by the lead instructor. / Aprendices: sección 2. Instructores: en custodia del instructor líder.
How rights were acquired / Cómo adquirió los derechos, si no es el autor	Not applicable: the rights holders are the authors / No aplica: los titulares son los autores
Document that proves it / Documento que lo prueba	Not applicable / No aplica

5. The work / La obra (art. 5, num. 4)

Field / Campo	Value / Valor
Title / Título de la obra	SomnGuard
Year of creation / Año de creación	2026
Country of origin / País de origen	Colombia
Version registered / Versión a registrar	`final-delivery-2026`
Short description of its functions / Breve descripción de sus funciones	Software que analiza señales visuales del conductor en el dispositivo edge para detectar indicios de somnolencia o fatiga, activar una alerta sonora local y sincronizar eventos y evidencias para consultarlos en el portal y la app.
What sets it apart from similar software / Qué la diferencia de obras similares	Procesa la detección y alerta crítica en el dispositivo, incluso sin conectividad; cuando recupera conexión, sincroniza los eventos con una plataforma cloud para su consulta y gestión.
Programming languages / Lenguajes	Java, TypeScript/JavaScript, Python y SQL.
Type: web, mobile, desktop, device / Tipo: web (API y portal), móvil y dispositivo edge.	

6. Declarations / Declaraciones (art. 4 y art. 5, num. 5 y 6)

Declaration / Declaración	Option / Opción
Origin / Origen	[x] Original [ ] Derived work / Obra derivada
If derived, base work / Si es derivada, obra de base	No aplica: SomnGuard se declara como obra original.
Type of work / Clase de obra	[ ] Individual [x] In collaboration / En colaboración [ ] Collective / Colectiva [ ] Anonymous / Anónima [ ] Pseudonymous / Seudónima [ ] Posthumous / Póstuma
Publication / Publicación	[ ] Unpublished / Inédita [x] Published / Publicada
If published, date and place / Si está publicada, fecha y lugar	2026-10-07, Hostinger; portal web desplegado y accesible públicamente: https://somnguard.tech/

Software is treated as unpublished unless the rights holder states otherwise (art. 4). A public repository may count as publication: the team records it here, and the Centro decides the final declaration.

El software se considera inédito salvo manifestación del titular (art. 4). Un repositorio público puede contar como publicación: el equipo lo deja anotado aquí y el Centro define la declaración final.

7. Elements attached / Elementos que se adjuntan (art. 2, 3 y 6)

At least one is required. Attach all three. / Se exige al menos uno. Adjunte los tres.

Element / Elemento	Where it is / Dónde está	Ready / Listo
Computer program / Programa de computador	ZIP files `final-delivery-2026` listed in `code-tag/README.md` / ZIP relacionados en `code-tag/README.md`	[x]
Program description / Descripción de programa	`02-software-analysis/` and `04-software-design/`	[x]
Auxiliary material / Material auxiliar	`08-manuals/`	[x]

8. Third-party components / Componentes de terceros

Inventory scope and evidence / Alcance y evidencia del inventario. On 2026-11-07, the five code repositories (`api`, `app`, `db`, `device`, and `portal`) were reviewed at their `final-delivery-2026` tags. The mobile app lockfile contains 945 versioned package entries and the portal lockfile contains 132; every versioned entry in both files has a license field. These totals include transitive, development, test, and platform-specific packages. The lockfile license fields are package metadata, not a legal review of each license or proof that every package is shipped in a production build.

Alcance y evidencia del inventario. Se revisaron los cinco repositorios de código (`api`, `app`, `db`, `device` y `portal`) en sus tags `final-delivery-2026`. El archivo de bloqueo de la app móvil contiene 945 paquetes con versión y el del portal 132; todos tienen un campo de licencia. Los conteos incluyen dependencias transitivas, de desarrollo, pruebas y plataformas. Esos campos son metadatos del paquete: no equivalen a una revisión legal individual ni demuestran que todos se incluyan en el artefacto de producción.

Component / Componente	Release evidence and version / Evidencia y versión	License and review status / Licencia y estado
Device image processing: `opencv-python`	`somnguard-device/pyproject.toml` declares `opencv-python>=4.8`; no Python lockfile is in the tag.	The Python wrapper is MIT; OpenCV itself is Apache-2.0. The standard wheel may also bundle FFmpeg (LGPL-2.1), and non-headless Linux wheels may bundle Qt (LGPL-3.0). The exact installed wheel and its notices cannot be determined from the tag. Package licensing details.
Device numerical and HTTP libraries: NumPy, HTTPX	The tag declares `numpy>=1.24` and `httpx>=0.25`; no resolved versions are recorded.	NumPy core: modified BSD / BSD-3-Clause; HTTPX: BSD-3-Clause. NumPy license, HTTPX source. Exact package versions and bundled binary notices remain unpinned.
Device vision framework: MediaPipe	The tag declares `mediapipe>=0.10`; no resolved version is recorded.	MediaPipe framework code: Apache-2.0. This is separate from the licenses of downloaded model files. Official repository.
Device optional dependency groups	`pyproject.toml` declares `dev` (pytest, pytest-asyncio, pytest-cov, Ruff, Black, mypy), `rpi` (Picamera2, RPi.GPIO), `full` (aiosqlite, Pydantic, pydantic-settings, simpleaudio), and `stream` (websockets, aiortc, LiveKit) extras.	These are optional and expressed as version ranges without a lockfile. The exact selected packages and their license notices depend on which extras are installed; resolve and record them for the release environment if they are part of the delivered device build.
Face Landmarker model bundle (`face_landmarker.task`)	Setup downloads Google's `face_landmarker/float16/1` bundle. A local, ignored copy was present on 2026-11-07: 3,758,596 bytes; SHA-256 `64184e229b263107bc2b804c6625db1341ff2bb731874b0bcc2fe6544e0bc9ff`. It is not in the Git tag.	Google's model cards identify BlazeFace, FaceLandmarker-V2, and Blendshape as Apache-2.0. Bundle and model cards, BlazeFace card.pdf), FaceLandmarker-V2 card, Blendshape card. Record the file in a controlled release artifact if that model copy is part of the delivery.
Phone detector model (`phone_detector.tflite`, EfficientDet-Lite0 float16)	Setup downloads Google's `efficientdet_lite0/float16/1` file. A local, ignored copy was present on 2026-11-07: 7,254,339 bytes; SHA-256 `4b59100025bea1235a84c1038879a6cccc9f6c49f5e41144e91e74d99e780993`. It is not in the Git tag.	Google's Object Detector guide identifies the model and COCO training data, but the reviewed official page does not state the exact downloaded model binary's license. Keep the model-specific redistribution/attribution terms pending; do not infer them from MediaPipe framework licensing.

Component / Componente	Release evidence and version / Evidencia y versión	License and review status / Licencia y estado
Mobile app direct libraries	`SomnGuard-app/package-lock.json` pins the graph. Representative direct runtime versions: React 19.2.3, React Native 0.86.3, Expo 57.0.27, Expo Router 57.0.25; LiveKit Client 2.22.3, LiveKit React Native 3.0.0 and Expo plugin 1.0.3; React Native WebRTC 144.2.0.	The lockfile reports MIT for React/React Native/Expo and the WebRTC package, and Apache-2.0 for the listed LiveKit packages. It also reports `lightningcss` under MPL-2.0, `caniuse-lite` under CC-BY-4.0, and `node-forge` under `(BSD-3-Clause OR GPL-2.0)`; verify whether these build/platform dependencies enter distributed artifacts and preserve applicable notices.
Web portal direct libraries	`package-lock.json` pins the graph. Representative runtime versions: React/React DOM 19.3.0, React Router DOM 7.18.4, LiveKit Client 2.22.3. Build/test tools include Vite 8.3.0, TypeScript 6.0.3 and Playwright 1.63.0.	Lockfile metadata reports MIT for React, React Router and Vite; Apache-2.0 for LiveKit Client, TypeScript and Playwright. The 132 versioned package records all contain license metadata; the lock includes MPL-2.0 packages among its build/platform dependencies.
API runtime libraries	`somnguard-api/pom.xml`, resolved with Spring Boot dependency BOM 4.1.1: Spring Boot 4.1.1; PostgreSQL JDBC 42.7.13; Liquibase Core 5.0.3; Springdoc 2.7.0; Micrometer Prometheus 1.17.1; Nimbus JOSE JWT 10.9.1; Google Auth 1.37.0; Logstash encoder 8.0; MinIO Java client 8.5.13.	Apache-2.0: Spring Boot, Springdoc, Micrometer, Nimbus JOSE JWT, Logstash encoder (8.0 license), MinIO Java client. BSD-2-Clause: PostgreSQL JDBC. BSD-3-Clause: Google Auth. **Liquibase Core 5.0.3 is FSL-1.1-ALv2**, not Apache-2.0. Versions for Spring Boot-managed artifacts come from the 4.1.1 BOM. The POM does not commit a resolved transitive Maven tree or generated third-party notices.
API tests and build toolchain	The POM marks H2, Spring Boot Test, Spring Security Test and Testcontainers as test scope; Testcontainers is declared as 1.20.4. The Dockerfile uses `maven:3.9-eclipse-temurin-21` and `eclipse-temurin:21-jre-alpine`.	Test-scope dependencies are not API runtime libraries. Docker base tags are not pinned to digests, so exact JDK, Maven, OS packages and their notices are not reproducible from the tag alone.
Database and supporting container images	`somnguard-db/docker-compose.yml` uses `postgres:16-alpine`; its Dockerfile uses `liquibase/liquibase:4.33`. API Compose uses `mailhog/mailhog:v1.0.1`, `adobe/s3mock:latest`, `livekit/livekit-server:v1.13`, and `coturn/coturn:4.6.2`.	PostgreSQL server: PostgreSQL License; Docker Official Image packaging: MIT. Liquibase **4.x image**: Apache-2.0; MailHog: MIT; S3Mock: Apache-2.0; LiveKit server: Apache-2.0; coturn: BSD-style 3-clause. `latest`, minor-only tags, and missing image digests/SBOMs prevent identifying every image layer and its full license set. PostgreSQL license, PostgreSQL image packaging, Liquibase 4 image license, S3Mock license, MailHog license, LiveKit license, coturn license.
External service integrations	API environment configuration includes Brevo email delivery and optional Firebase Cloud Messaging; LiveKit server can be enabled for streaming.	These are external services rather than redistributable code packages. Their provider terms, account permissions, and privacy obligations must be checked separately if enabled for the release; they are not represented as open-source package licenses here.
Project images and external media	The tagged app includes `SomnGuard-app/assets/images/somnguard-logo.jpg`, `icon.png`, `favicon.png`, `splash-icon.png` and React sample images; the portal includes `src/assets/hero.png`, `favicon.svg`, `icons.svg`, `react.svg` and `vite.svg`. Links to the pitch/demo videos are listed in `10-videos/videos.md`; source media is not tracked in the five code repositories.	The reviewed tags do not document the creator, source, or license/permission for each image or video element. Confirm team-created work or record the source/license/permission for any third-party logo, image, font, music, footage, or generated media before signing the originality declaration.

The deployment procedure also mentions SeaweedFS and Traefik, but the reviewed tagged repositories do not contain the VPS Compose file or image digests needed to confirm whether those images are part of the delivered deployment. No training dataset or separate audio file was found in the five tagged repositories; this does not establish the provenance of material stored outside Git.

El procedimiento de despliegue también menciona SeaweedFS y Traefik, pero los repositorios revisados no contienen el Compose del VPS ni los digests de imagen para confirmar si forman parte del despliegue entregado. No se encontró un dataset de entrenamiento ni archivos de audio separados en los cinco tags; esto no determina el origen de material guardado fuera de Git.

9. Private data / Datos privados

Do not write this data in this repository. Registration also asks for address, nationality, phone and email. Each member gives them to the instructor through the private channel the instructor indicates.

No escriba estos datos en este repositorio. El registro también pide dirección, nacionalidad, teléfono y correo. Cada integrante se los entrega al instructor por el canal privado que se indique.

10. Team declaration / Declaración del equipo

- [ ] The information is true and the names match the official ID documents. / La información es veraz y los nombres coinciden con los documentos de identidad.
- [ ] The listed authors are the people who created the software. / Los autores relacionados son quienes crearon el software.
- [ ] Third-party components are declared. / Los componentes de terceros están declarados.
- [ ] No passwords, tokens or keys are included. / No se incluyen contraseñas, tokens ni llaves.